# How excitation and inhibition are organised into task modules

Four budget points of the cog20 front, $W_{out}$ trained, 150k steps

canonical_mc

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This note explains one analysis step by step: we take Dale networks trained on the 20 Yang et al. (2019) tasks under different cost budgets, find groups of neurons that are involved in the same tasks, and ask how the inhibitory groups relate to the excitatory ones. Lesion experiments are left out here. The code is `notebooks/ei_modularity_150k.ipynb`; the figures are produced by `reports/ei_modularity/make_figures.py`, which repeats the same computation without the lesions.

## 1 The four networks

All networks have 256 units, 204 excitatory (E) and 52 inhibitory (I), and obey Dale's law: every outgoing weight of an E unit is positive and every outgoing weight of an I unit is negative. They were trained for 150 000 steps on all 20 tasks at once. Two costs can be limited during training:

- a **rate budget**, a ceiling on how much the units fire (metabolic cost);
- a **wiring budget**, a ceiling on the total recurrent weight (wiring cost). Because it is an L1 ceiling, keeping under it removes most synapses entirely.

We look at four points (Fig. 1a), each trained from 5 random seeds:

**control** no budget at all.

**A3 (rate only)** a tight rate budget, no wiring budget.

**B0 (wiring only)** a tight wiring budget, no rate budget.

**C0 (both)** both budgets tight at the same time.

Figure 1b shows what the trained networks actually spend. Each budget does what it is meant to: A3 fires about 250 times less than the control, B0 and C0 use about 6 times less wiring. The constrained networks are also very different in how many synapses they keep: control and A3 keep about two thirds of all possible synapses, B0 keeps 6 %, and C0 less than 1 %.

All four solve the tasks: the average accuracy over the 20 tasks is above 0.95 for every network. The constrained points pay for their budget on the hardest task, which is usually DNMS: the worst task is at 0.96 in the control and between 0.85 and 0.88 on average at A3, B0 and C0.

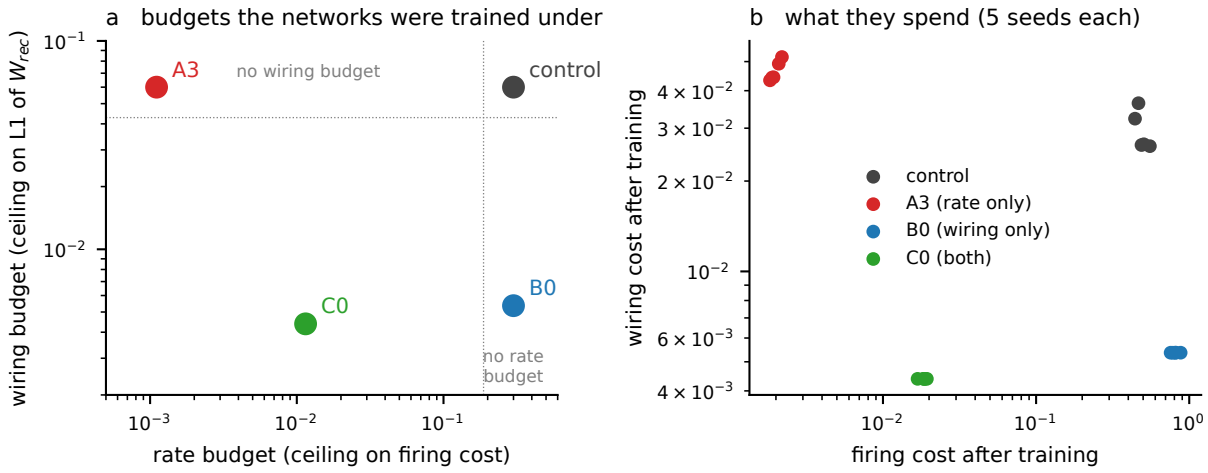



**Figure 1:** **a**, the budgets of the four points (log axes; points drawn outside the dotted lines have no budget on that axis). **b**, firing and wiring cost of the trained networks, one dot per seed.

## 2 Finding task groups, separately for E and for I

**Task variance.** For every unit and every task we measure how much the unit's activity varies across trials of that task (variance across trials at each time point, averaged over the trial), as in Yang et al. (2019). A unit with high task variance on a task carries information about that task. This gives a table of 20 tasks × 256 units. Units that barely vary on any task are dropped, and each remaining unit's 20 values are divided by its own maximum, so every unit has a *task profile* between 0 and 1 whose largest entry is 1.

**Clustering.** Yang et al. cluster all units together. Here E and I units are clustered *separately*: k-means on the task profiles of the E units, and independently on those of the I units. The number of groups is chosen from the silhouette score, as Yang does. The result is a set of E groups and a set of I groups, each with an average task profile. The top panels of Figs. 4–7 show these groups: each column is a unit, each row a task, and white means the unit varies most on that task.

**How broad are the neurons?** A simple number per population is the *number of tasks per neuron*: for how many of the 20 tasks a unit's profile is above half of its maximum. A unit that cares about one task scores 1, a unit that is involved in everything scores 20.

## 3 Three ways E and I could be organised

Once we have E groups (modules), there are three simple possibilities for the inhibitory neurons (Fig. 2).

**Type 1: E modules with shared inhibition.** The task modules are purely excitatory. Inhibition is a common pool that is active in all tasks and inhibits everyone, for instance to keep overall activity in check. I neurons should then be broader than E neurons, and their connections should not care which task an E group does.

**Type 2: mixed E+I modules.** Each module has its own excitatory and inhibitory neurons. An I group is tuned to the same tasks as one E group and is wired mainly to that E group, in both directions: the E group drives its I group, and the I group inhibits its own E group (local feedback inhibition).

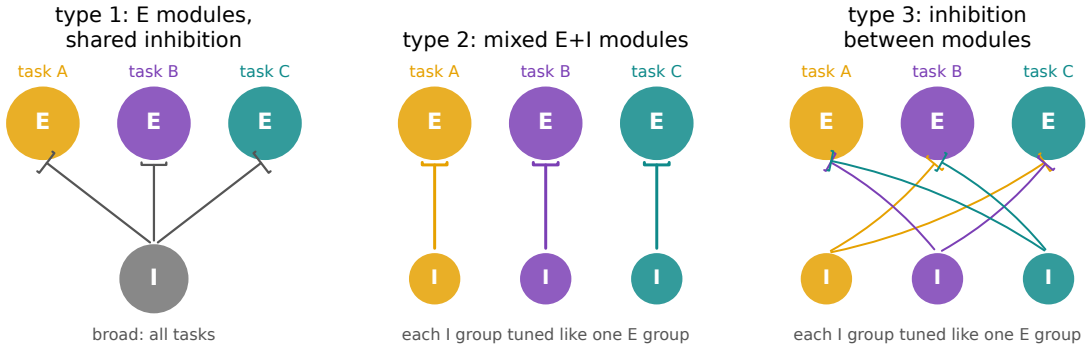



**Figure 2:** The three kinds of E/I organisation. Circles are groups of neurons, coloured by the task they are involved in; bars are inhibitory connections.

**Type 3: inhibition between modules.** I groups are also tuned like E groups, but they inhibit the *other* modules. When one module is active it suppresses the competing ones (lateral inhibition).

Types 2 and 3 look the same when we only look at tuning: in both, every I group has an E group with the same task profile. They differ only in the wiring, and that is what the next section measures.

## 4 Measuring whether the wiring follows the tasks

For one network we compute two tables with one entry per pair of an I group and an E group (Fig. 3, using C0 as the example):

1. **Task similarity** (panel a): the correlation $r$ between the I group's and the E group's average task profile. $r$ near 1 means the two groups are involved in the same tasks.
2. **Wiring strength** (panel b): the average absolute weight from every unit of the I group to every unit of the E group (synapses that were pruned count as 0). The raw number depends on group sizes and on how many synapses the network has overall, so we compare it with the same number after shuffling which unit belongs to which group (500 shuffles, group sizes kept) and report how many standard deviations it lies above the shuffles ($z$). $z = 0$ means the two groups are as connected as two random sets of neurons of the same size.

Then we ask whether the two tables go together (panel c): the Spearman correlation $\rho$ between task similarity and wiring strength over all (I group, E group) pairs of the network.

- $\rho > 0$: I groups are wired most strongly to the E groups that do the same tasks $\rightarrow$ type 2.
- $\rho < 0$: I groups are wired most strongly to E groups that do *different* tasks $\rightarrow$ type 3.
- $\rho \approx 0$: wiring ignores the tasks $\rightarrow$ type 1.

The same is done for the opposite direction, E $\rightarrow$ I. Type 2 predicts positive $\rho$ in both directions.

## 5 Results

For each point we show the seed with the best worst-task accuracy, and then check against the other four seeds (Fig. 8).

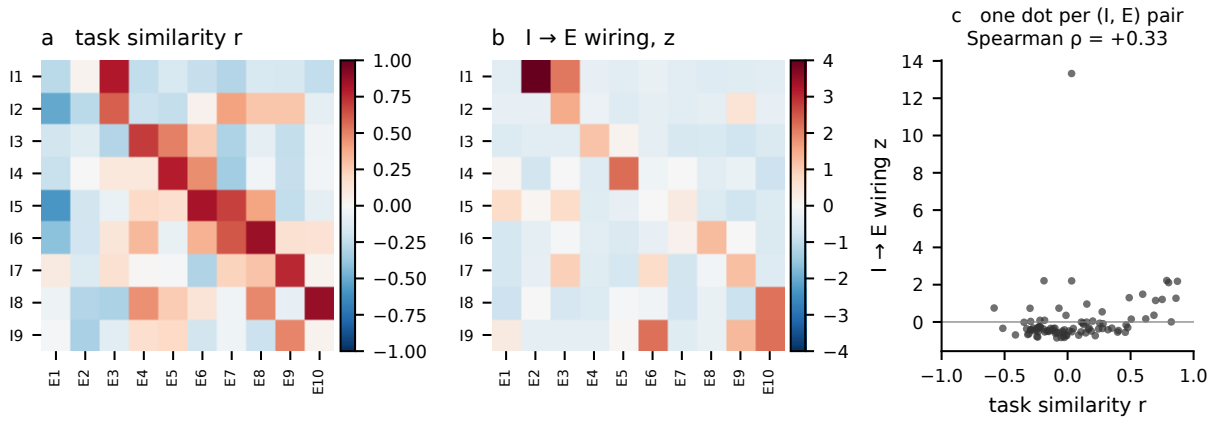



**Figure 3:** The wiring measure for one network (C0, seed 2). **a**, task similarity of every I group (rows) with every E group (columns). **b**, I → E wiring strength as a $z$-score against shuffled groups. **c**, the two tables plotted against each other: one dot per (I group, E group) pair. Here the pairs that share tasks (right) tend to be more strongly connected, $\rho = +0.33$. One pair (I1 → E2) is far stronger than all others: a small I group projecting to the RT Anti group.

## Control: inhibition between modules (type 3)

The unconstrained network has few, broad E groups: about 5 tasks per neuron, and the groups are task *families* (all DM tasks, all delayed DM tasks, the matching tasks) rather than single tasks (Fig. 4). The I neurons are tuned as narrowly as the E neurons, and every I group has an E group with a similar profile.

The wiring shows the clearest effect of the four points: I → E $\rho$ is negative in all five seeds (between $-0.3$ and $-0.7$). An I group inhibits the E groups of *other* tasks more than chance, and its own E partner less. The E → I direction is unstructured ($\rho$ around 0). This is lateral inhibition between modules.

Note that the control keeps two thirds of all synapses, so here the measure is about how *strong* the synapses are, not about whether they exist.

## A3, rate only: sharp E modules, wiring ignores tasks

Limiting firing gives the most specialised E neurons of all four points: about 3 tasks per neuron, and many small E groups, several of them dedicated to a single task (RT Go, RT Anti, DMS, DNMS, DMC; Fig. 5). The I neurons do not follow this: in every seed they are broader than the E neurons (about 4 tasks per neuron) and fall into fewer groups. Their profiles still resemble an E group, but at the level of task families (for example all delayed tasks) rather than single tasks.

The wiring does not follow the tasks at all: $\rho$ is within $\pm 0.06$ of zero for I → E in every seed, and around zero for E → I. Like the control, A3 keeps most synapses, so this is again about weight strength.

A3 is therefore closest to type 1: the excitatory network is modular, while inhibition is broader and not wired to particular modules.

## B0, wiring only: weak, seed-dependent E+I pairing

With only a wiring budget the E neurons are the broadest of the four points (about 5 tasks per neuron), and the E groups are few and large (four in the seed shown, Fig. 6). The task structure is the weakest here.

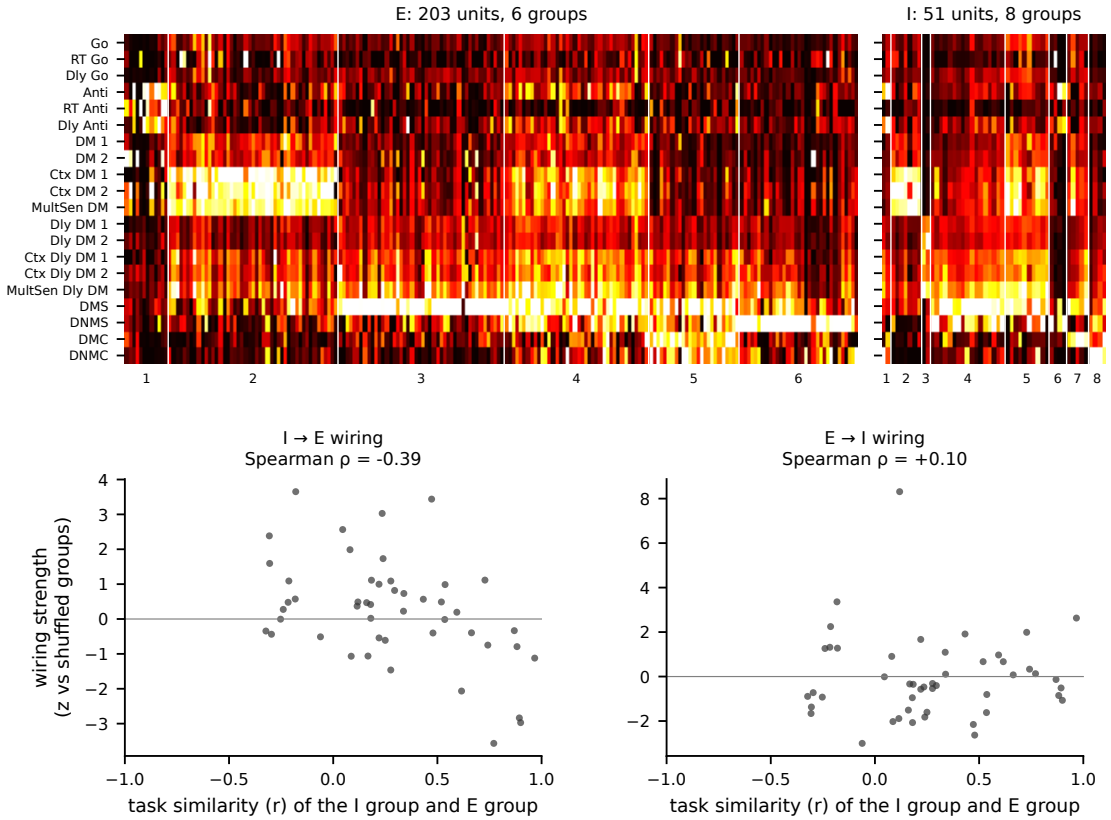



**Figure 4:** Control, seed 1. Top: normalised task variance of the active E units (left) and I units (right), units sorted by group. Bottom: wiring strength against task similarity for every (I group, E group) pair, for I → E and E → I.

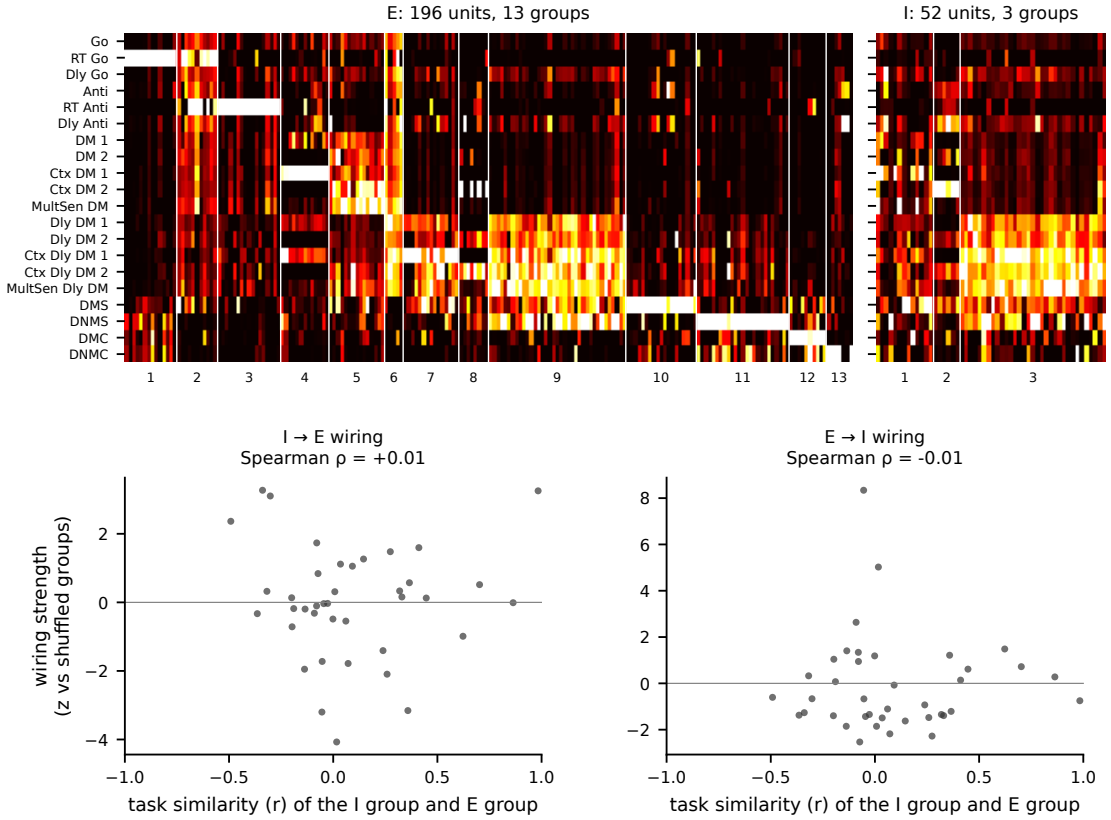



**Figure 5:** A3 (rate only), seed 3. Same layout as Fig. 4.

The wiring tends to be positive in both directions, $\rho \approx +0.2$ on average, but the seeds disagree: only two of the five seeds are clearly positive in both directions, the others are close to zero in at least one. This is a weak hint of type 2, not a clear result. Unlike the control and A3, only 6 % of synapses remain, so here the measure also reflects which synapses exist.

### C0, both budgets: mixed E+I modules (type 2)

With both budgets the E neurons are sharply tuned (about 3.5 tasks per neuron) and form many groups, close to A3. Unlike A3, the I neurons are only slightly broader than the E neurons (about 4 tasks per neuron) and form more groups (Fig. 7).

Here the wiring follows the tasks. E $\rightarrow$ I $\rho$ is clearly positive in all five seeds (between +0.3 and +0.6): an E group connects mostly to the I groups that are involved in the same tasks. The I $\rightarrow$ E direction is positive in all five seeds as well, but weaker (+0.1 to +0.3). With less than 1 % of synapses left, this is a statement about which connections the network kept. C0 is the only point that matches type 2 in both tuning and wiring.

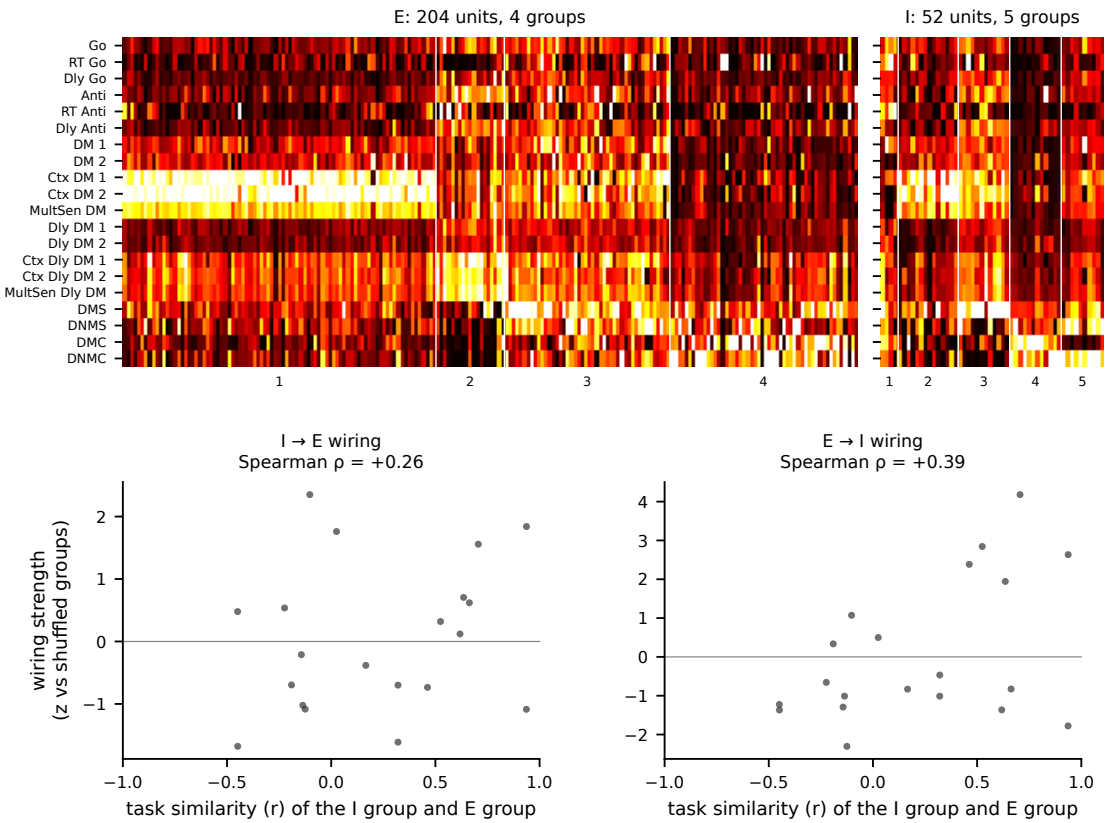



**Figure 6:** B0 (wiring only), seed 0. Same layout as Fig. 4.

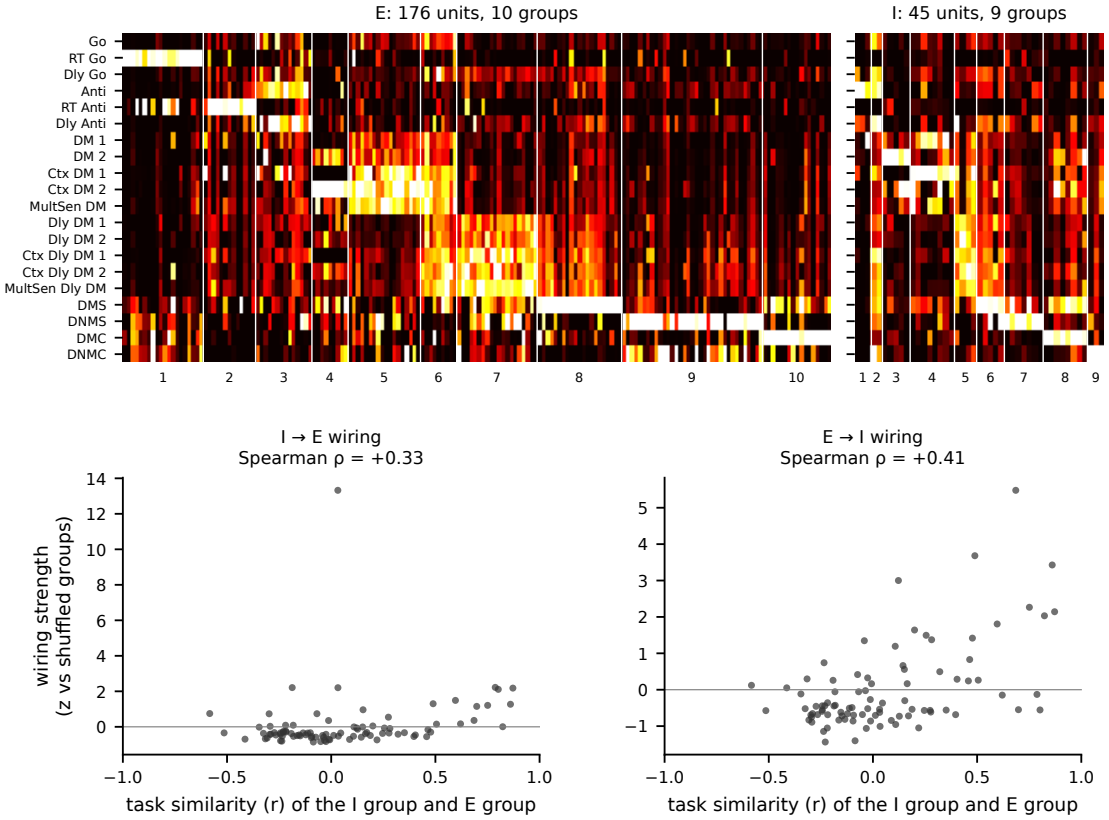



**Figure 7:** C0 (both budgets), seed 2. Same layout as Fig. 4.

# 6 Summary

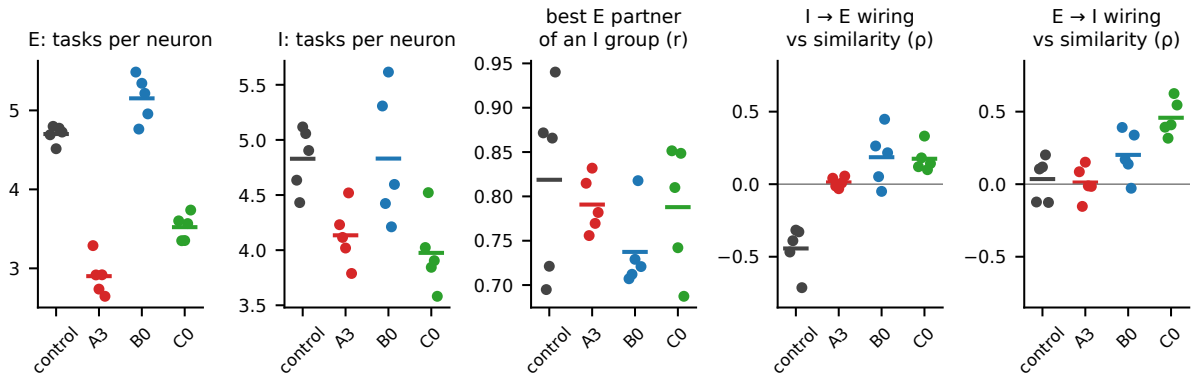



**Figure 8:** The five numbers of this note for all 20 networks. Dots are seeds, bars are means over the 5 seeds.

| | control | A3 (rate) | B0 (wiring) | C0 (both) |
|---|---|---|---|---|
| E tasks per neuron | 4.7 | 2.9 | 5.2 | 3.5 |
| I tasks per neuron | 4.8 | 4.1 | 4.8 | 4.0 |
| I → E wiring vs task similarity ($\rho$) | −0.44 | 0.01 | 0.19 | 0.18 |
| E → I wiring vs task similarity ($\rho$) | 0.04 | 0.01 | 0.20 | 0.46 |
| synapses kept | 70 % | 61 % | 6 % | 0.7 % |
| closest type | 3 | 1 | weak 2 | 2 |

**Table 1:** Means over 5 seeds.

Read together:

- Without budgets, the network forms broad modules that **inhibit each other** (type 3).
- A **rate budget** sharpens the excitatory modules, but inhibition stays broader and is not wired to them (closest to type 1).
- A **wiring budget** alone makes neurons broad and gives at most a weak E+I pairing.
- **Both budgets together** give sharp excitatory modules **with their own inhibitory neurons**, wired to them mainly in the E → I direction (type 2).

Every I group in every network has an E group with a similar task profile (best match $r \approx 0.7$–$0.9$ everywhere), so the four points differ in the breadth of I tuning and above all in the wiring, not in whether the I neurons are tuned at all.

**Caveats.** Five seeds per point. Each network's $\rho$ is computed from its own groups, between about 12 and 170 (I, E) pairs, and the number of groups changes from seed to seed. We only make claims where the seeds barely overlap: the negative I → E $\rho$ of the control, the positive E → I $\rho$ of C0, the zero wiring $\rho$ and broader I neurons of A3. In the control and A3 almost every synapse exists, so their wiring numbers describe weight strength only. Lesion experiments, which test whether the I part of a module is actually needed, are in the notebook and not discussed here.